

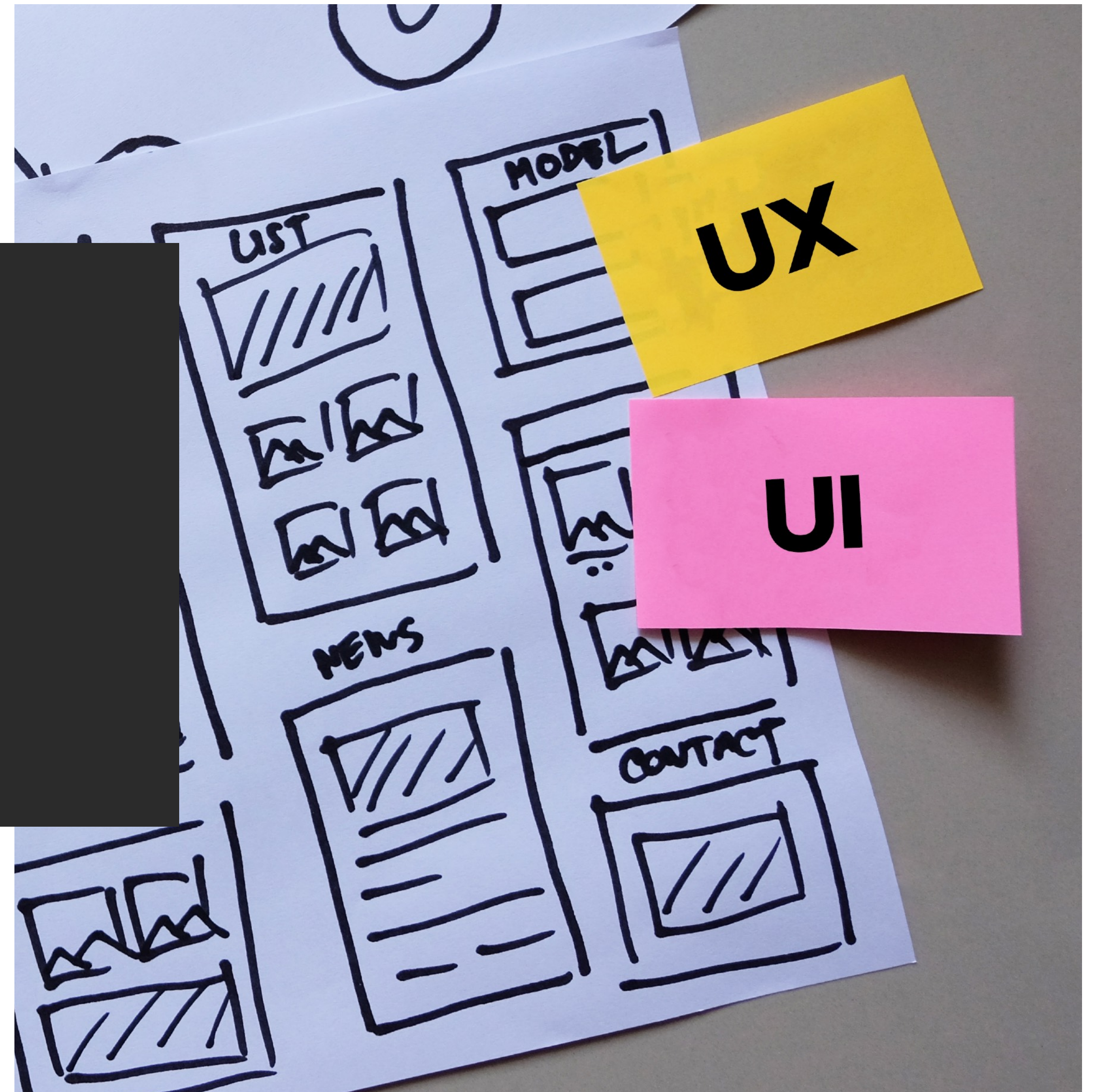
Interface Design

Ing. Lisa-Maria Freller, MA

Design Patterns

Overview

10/4 · 17/4 Design Patterns
8/4 · 7/4 Interaction & AI
29/5 · 12/6 Design Systems
26/6 Responsive Design



Today

- 📌 Figma
Editor, Frames, Auto-Layout, Layers, Assets, Pages, Text, Components
- 📌 Design Patterns
Definition, Classification & Best Practices
- 📌 Dark Patterns
- 📌 UX Laws & Persuasive design

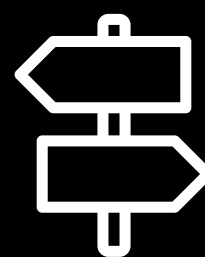


🎯 Figma refresh & apply Design Patterns

Figma



inola



UX/UI Designer · FE-Developer
industrial background

Until 2023 Research assistant · industry
experience

Since 2023 Part-time lecturer

Since 2024 UX/UI Designer @ inola

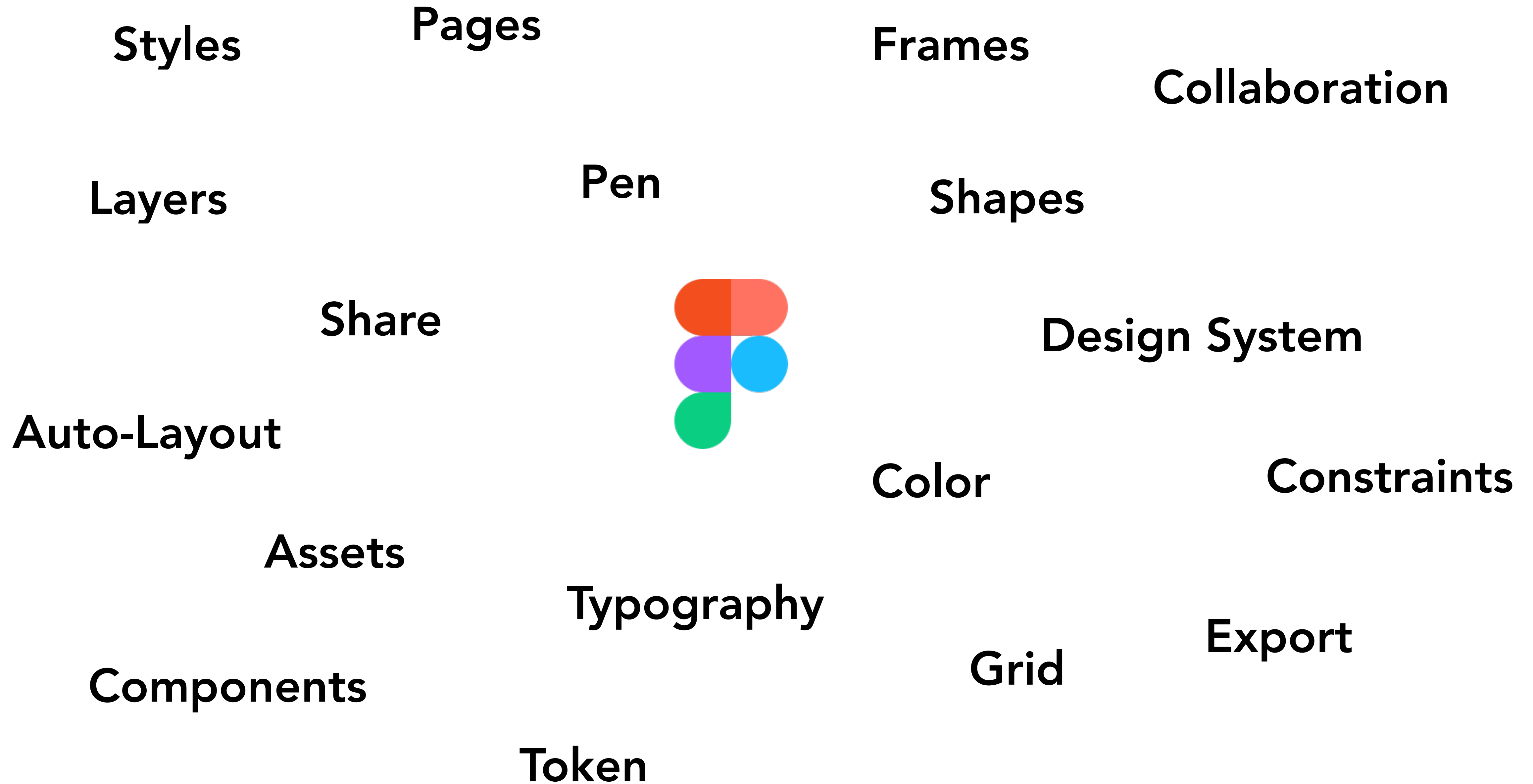
HTL Leonding · Computer Science & Media
Technology

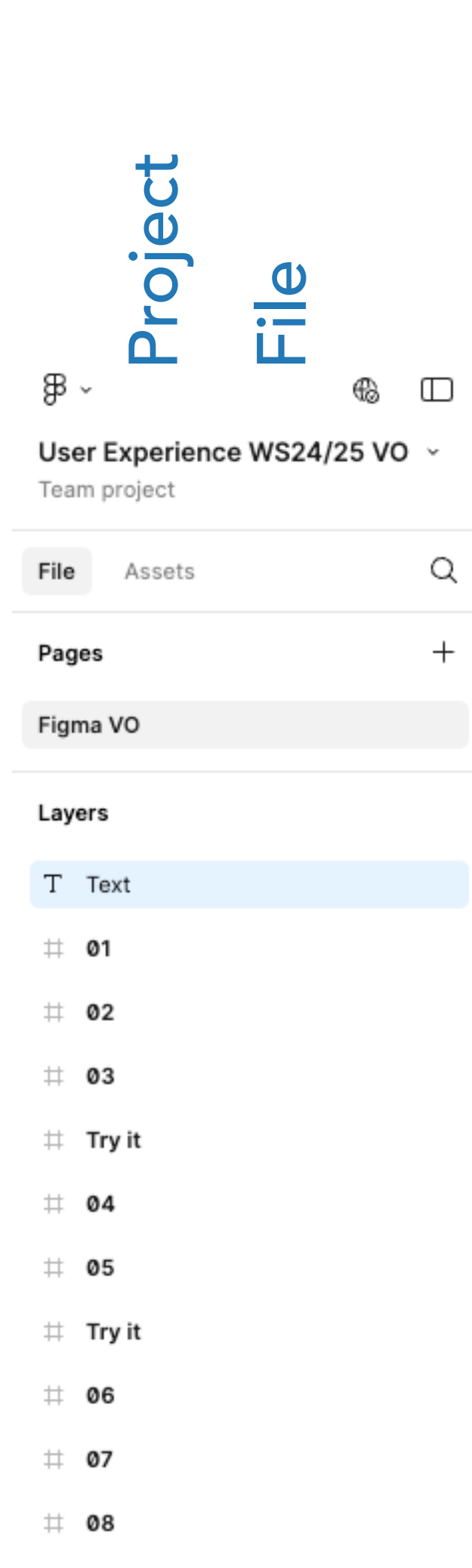
BA & MA @ Hagenberg · Communication &
Knowledge Media

Figma

- ✓ Web-based
- ✓ Vector-based
- ✓ 2019 · Figma launched Figma Community, allowing designers to publish work
- ✓ 2021 · Figma launched a digital whiteboarding tool (FigJam)
- ✓ 2025 · Focus on AI automating repetitive tasks & enhancing creativity (FigmaMake), better integration in development flows and interactivity

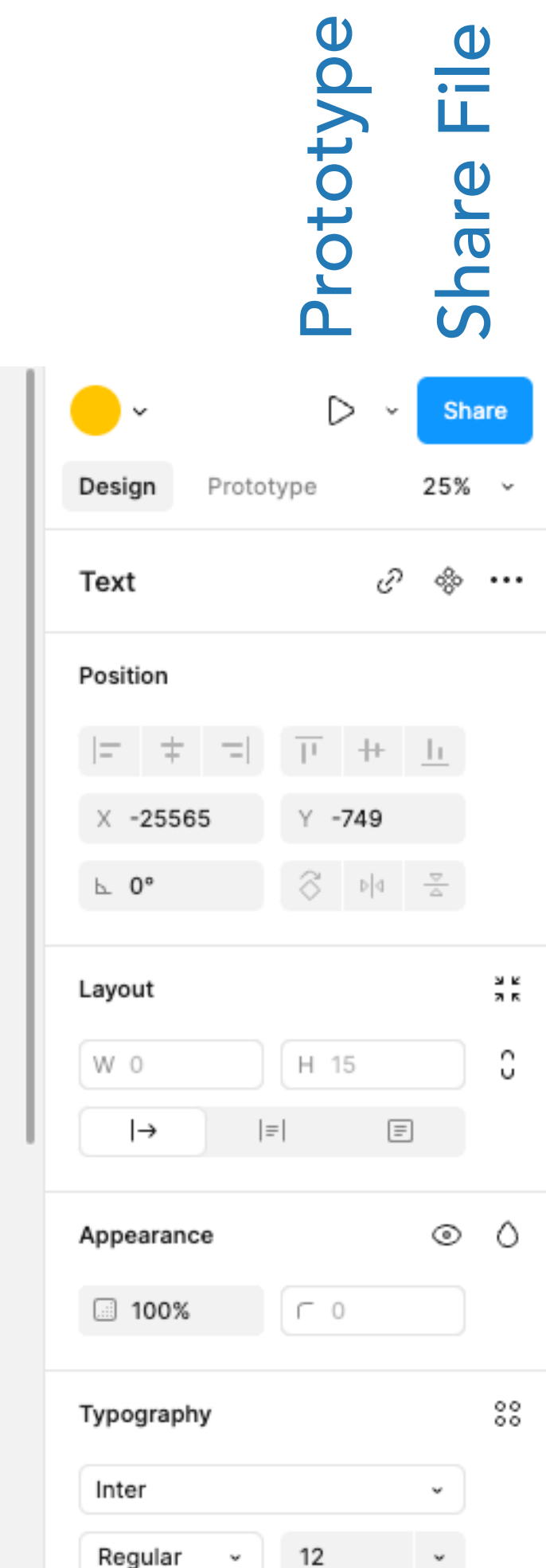
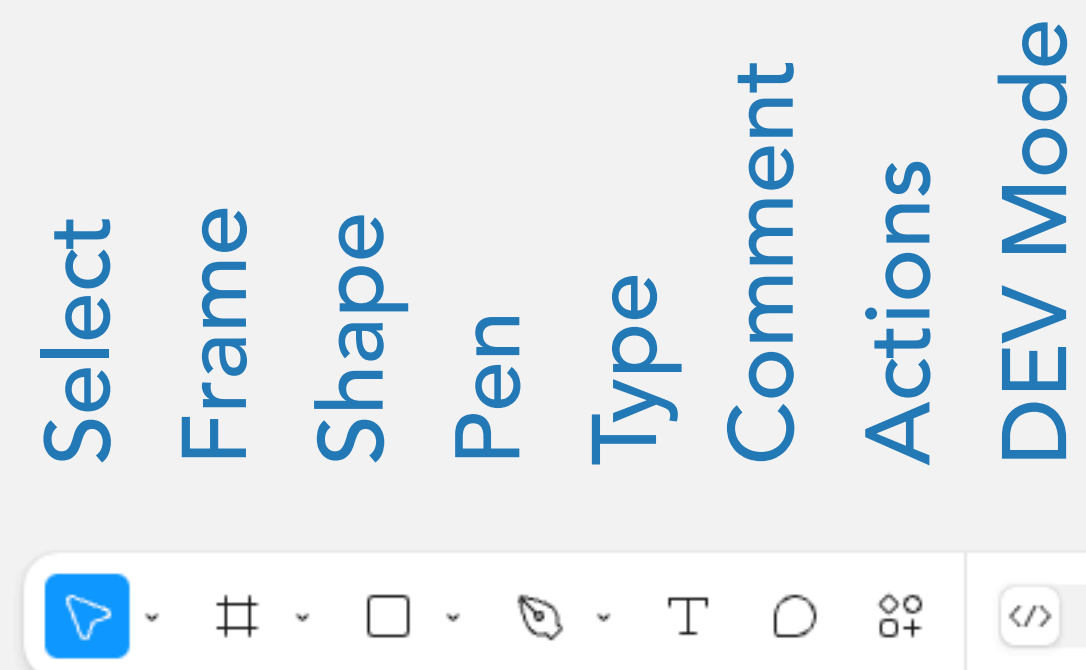






Layers Panel

✓ layers, assets, pages



Properties Panel

✓ customise frame size, position, color, stroke, effects

✓ view element „code“

Frames

Like artboards or containers for elements

- Screen sizes like Android, iPhone ..
- Needed for interactive prototypes

✓ Use frames instead of groups

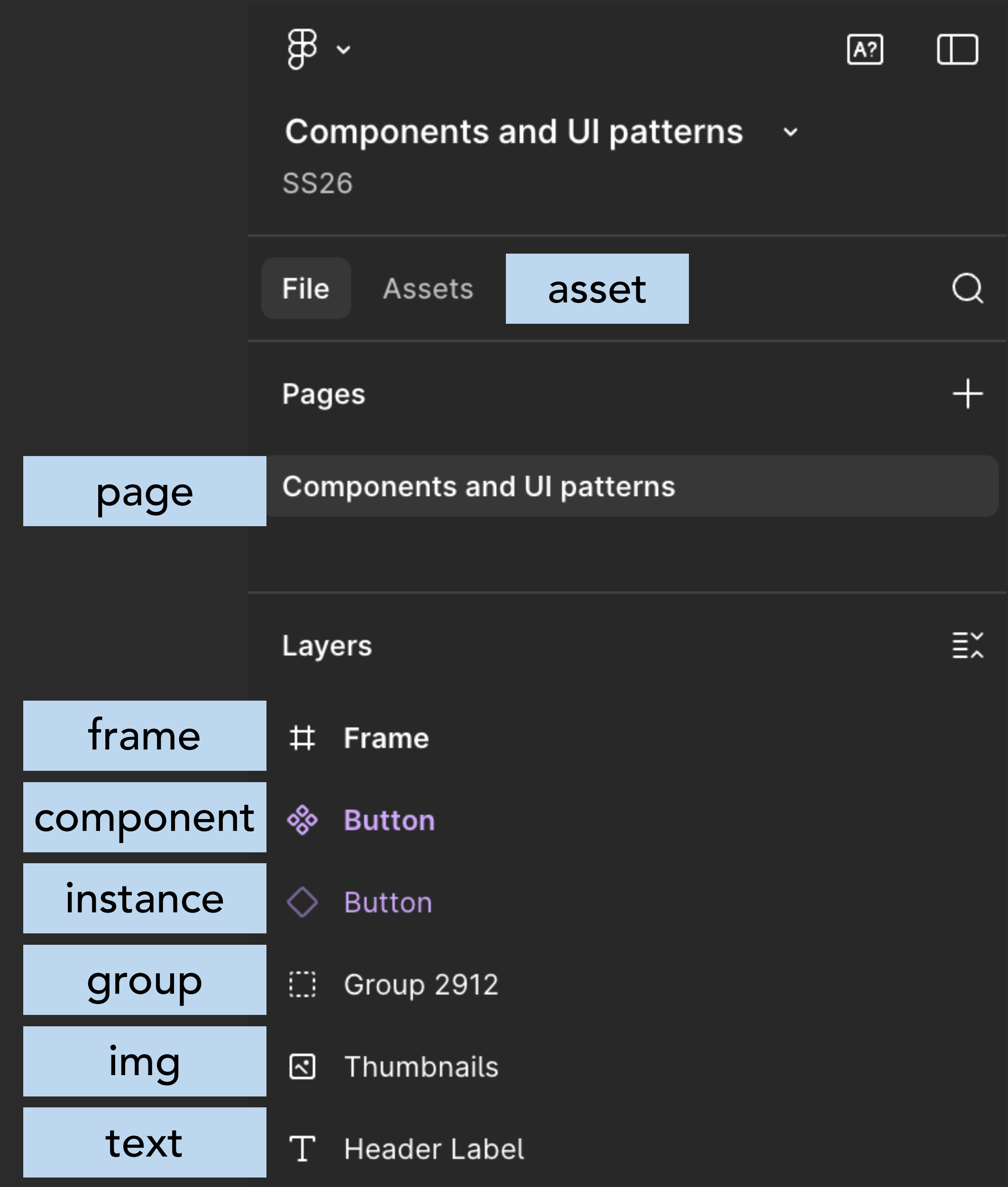


Design	Prototype	Inspect
Frame		
▼ Phone		
iPhone 14		390 × 844
iPhone 14 Pro		393 × 852
iPhone 14 Plus		428 × 926
iPhone 14 Pro Max		430 × 932
iPhone 13 Pro Max		428 × 926
iPhone 13 / 13 Pro		390 × 844
iPhone 13 mini		375 × 812
iPhone SE		320 × 568
iPhone 8 Plus		414 × 736
iPhone 8		375 × 667
Android Small		360 × 640
Android Large		360 × 800

Layers, Assets, Pages

Each new element automatically creates layer

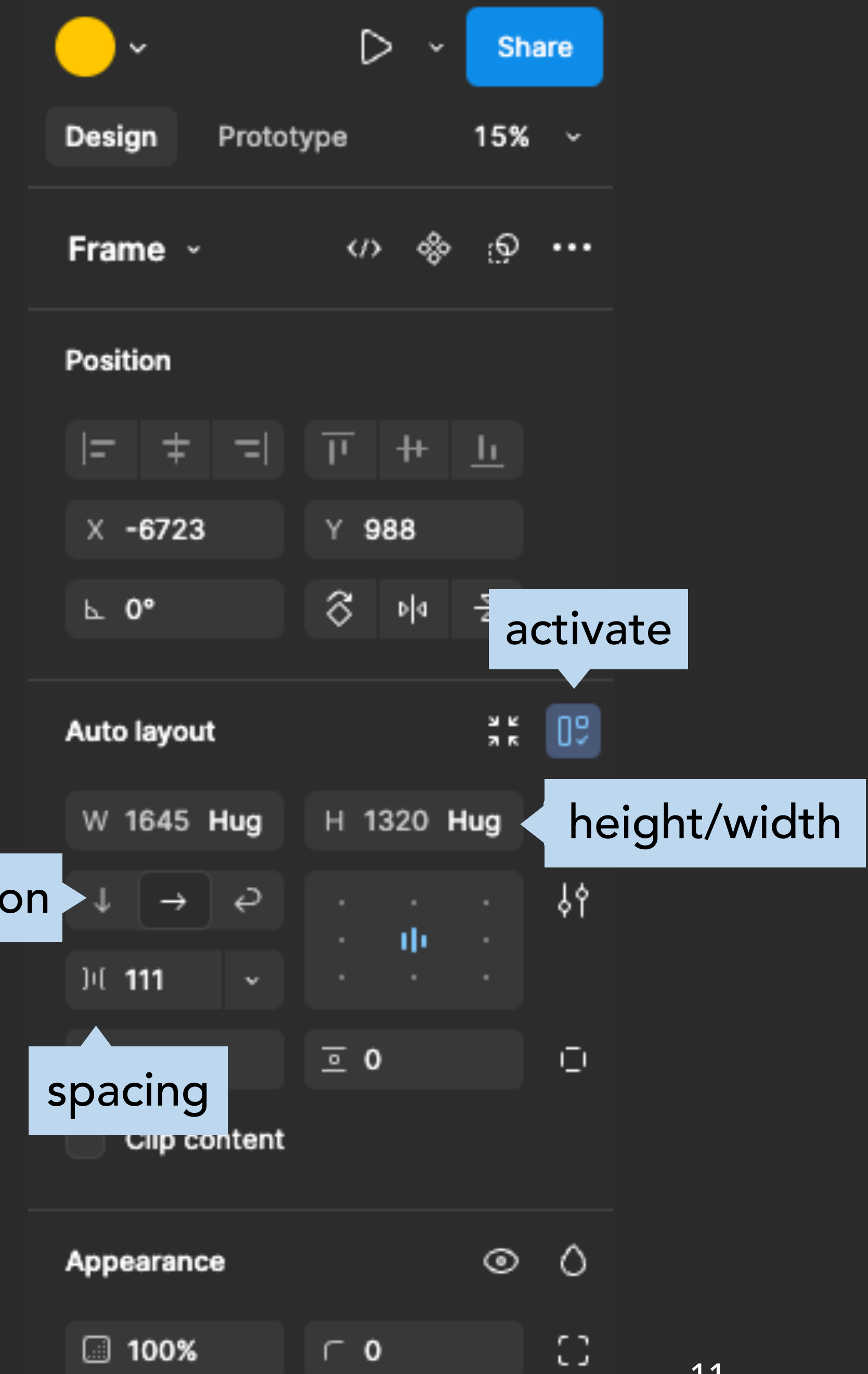
- **Group:** select layer and click CMD+G to keep file organised, select an item within a group: hold CMD
- **Page:** set up different sections or areas e.g. Brainstorm, Wireframes,...
- **Asset:** store components



Auto-Layout

Dynamic frames that respond to their content

- Frames or components **grow/shrink** based on content
- Auto-layout is a child layer inside a parent frame
- Shift+A or 
- Select vertical or horizontal **direction** and **spacing** in the right auto-layout menu
- fix or max/min height/width

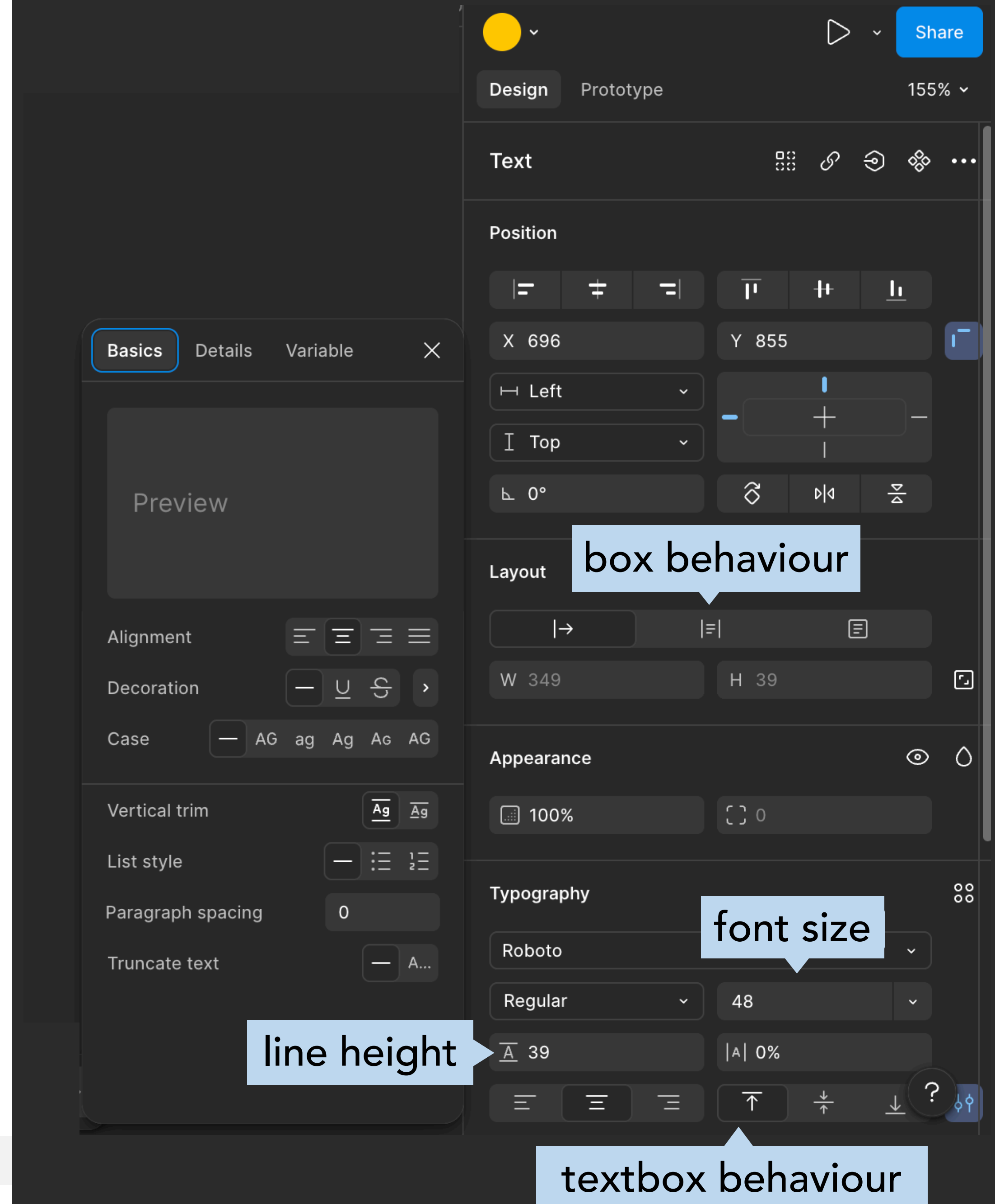


Textbox

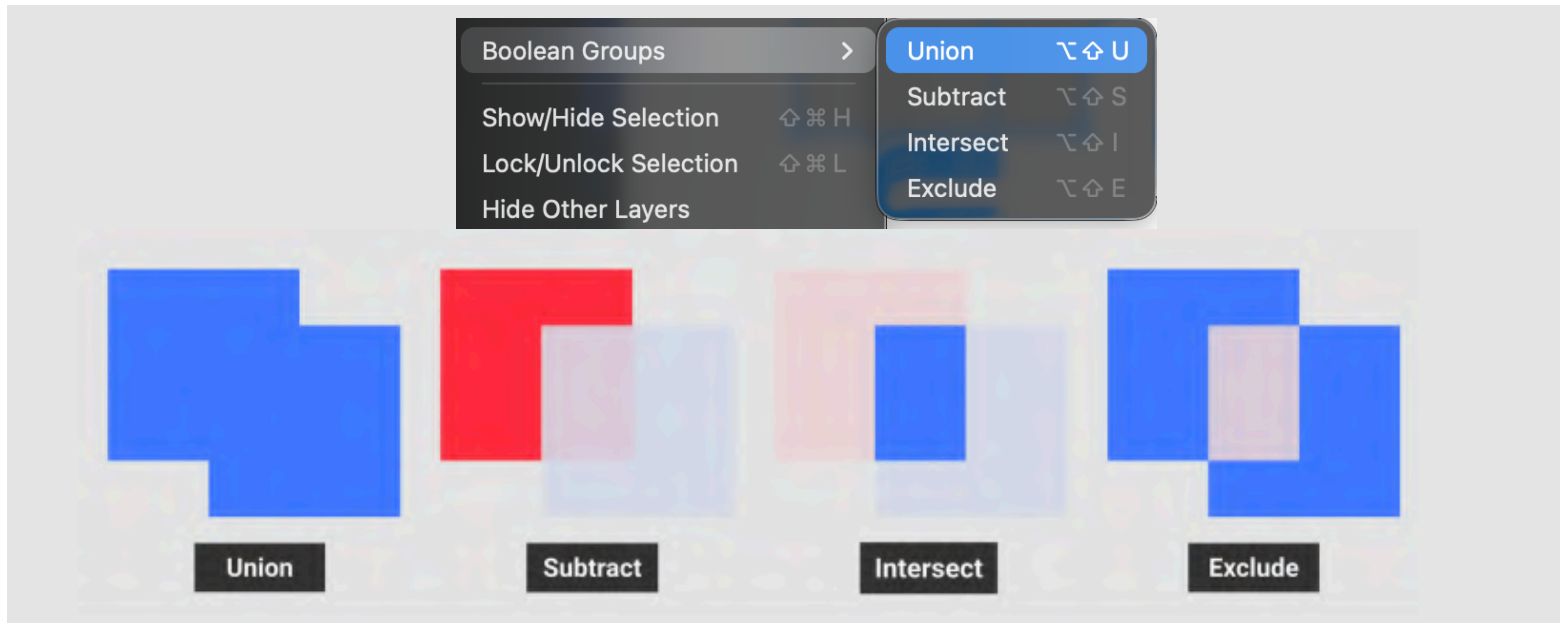
Place Textboxes with T

Configuration is important for Auto-Layout and later size adaption

✓ Text should not be smaller than 12-14px

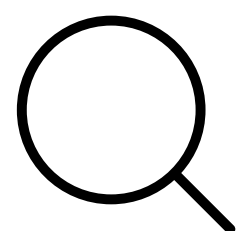


Shapes

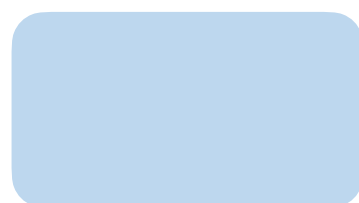


Atomic Design

Atom



Icon



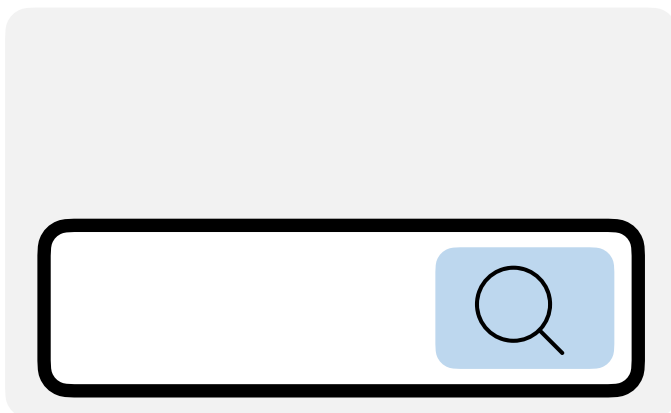
Button

Molecule



Search Bar

Organism



Card with Search Bar

Components

The building blocks of any website

Four main aspects to consider when designing components:

- ✓ **Usability**: How easy it is to use the component
- ✓ **Aesthetics**: The visual appearance
- ✓ **Functionality**: The way the component works
- ✓ **Accessibility**: The ability for everyone to use it



Components

Reusable design part eg. Button

- ❖ **Main component** defines properties & styles
- ❖ **Instance** is an identical copy, can be found in assets tab and placed in your design

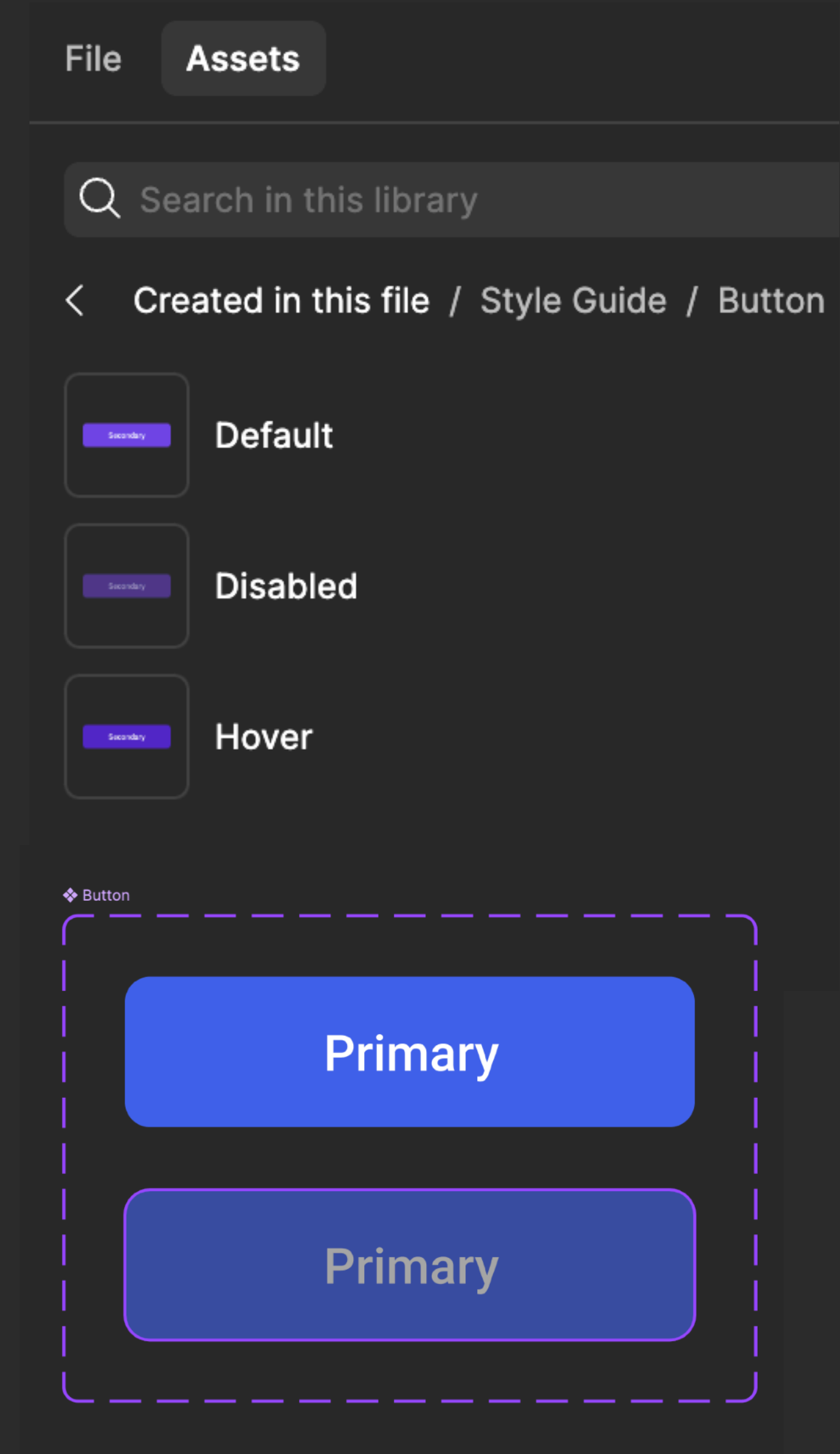
Updating the main component also updates all instances

Copy	⌘ C
Paste to replace	⌘⇧⌘ V
Copy/Paste as	▶
Select layer	▶
Move to page	▶
Bring to front	]
Send to back	[
Convert to section	
Group selection	⌘ G
Frame selection	⌘⇧⌘ G
Ungroup	⌘⇧⌘ ⌫
Flatten	⌘ E
Outline stroke	⇧⌘ O
Use as mask	^⌘ M
Add auto layout	⇧⌘ A
Create component	⌘⇧⌘ K

Components

Components should be build on Frontend Framework design

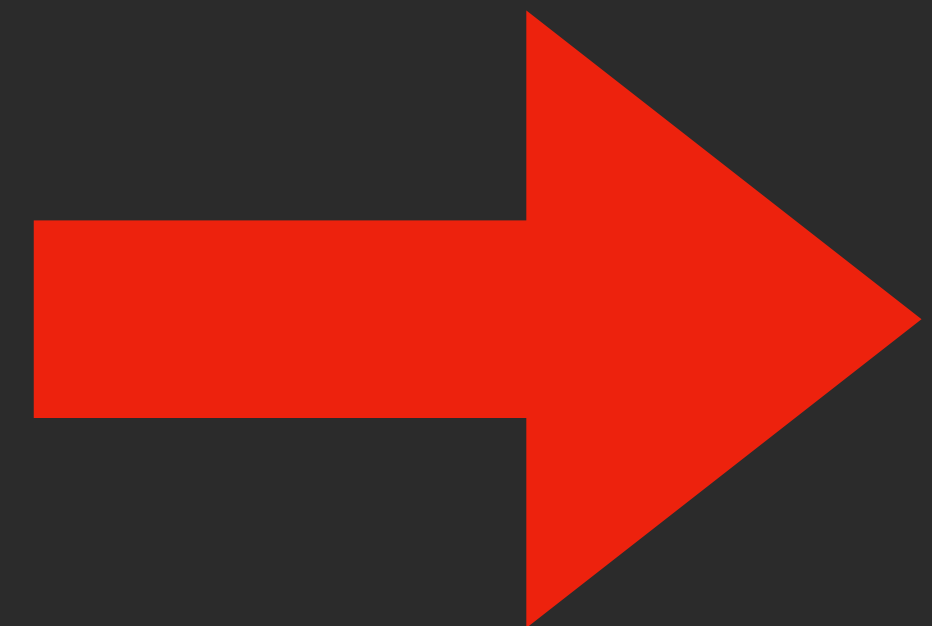
- ✓ Components can have **properties** for texts, displays,..
- ✓ Create **variants** for eg. button states (normal, pressed,..)
- ✓ Nesting possible e.g icon inside a button
- ✓ Automatic **folders**: name them e.g. icon/user, icon/home



👋 Hands on

- ✓ Create standard shapes: press R (Rectangle) L (Line) O (Ellipse)
- ✓ Shift+key keeps proportions
- ✓ Complex shape: press P

Create a complex shape



👋 Hands on

Use a Frame as a base, then create a **Button Component**

The design should clearly resemble a CTA clickable button

Requirements

- ✓ Text label
- ✓ Include your shape and some icons eg. Star as nested components
the icon should be visually aligned
- ✓ Variants (default / hover / disabled / pressed)
- ✓ Sizes (S / M / L)

Button styling, apply

- ✓ Background color
- ✓ Rounded corners
- ✓ Proper padding
- ✓ Stroke or shadow

X DON'T group elements instead of Auto Layout

X NO inconsistent spacing or misalignment

👋 Hands on

Design a reusable
Job Listing Card Component
for a Job Platform

Variants

- ✓ Default
- ✓ Featured Job
- ✓ Saved Job
- ✓ Compact Version

Requirements

- ✓ Company Logo + Name
- ✓ Job Title + Location
- ✓ Income in € (optional)
- ✓ Components:
 - ✓ Job-Type Chip (Full-time / Remote)
 - ✓ Your Button Component as „Apply“-Button Instance
 - ✓ Bookmark Icon

Complexity

- ✓ Auto Layout
- ✓ Boolean properties (e.g. show Salary / hide Salary)
- ✓ Nested components (Button and Chip)

Design Patterns

Design Patterns

A UI pattern is a recurring digital solution to a interface problem in a given context

Patterns ensure:

- ✓ Consistency
- ✓ Predictability
- ✓ Familiar interactions
- ✓ Evidence to support a solution



Pattern Classification

Classifying the UI patterns by **problem solved** helps designers:

- ✓ Understand user needs
- ✓ Choose proven design solutions
- ✓ Design interfaces faster
- ✓ Consistency

System Response

Inform users about what the system is doing or what happened after an action

Examples:

- Notifications
- Alerts
- Progress indicators

Input & Interaction

Help users enter information or interact with the system efficiently

Examples:

- Login forms
- Validation feedback
- Autocomplete fields

Pattern Classification

Navigation 🙄

Help users move through information and find content

Examples:

- Tabs
- Pagination
- Infinite Scroll
- Search Filter

Discovery 🔍

Help users explore, browse, or discover content

Examples:

- Thumbnails
- Activity feeds
- Favourites / wish lists
- Autocomplete search

Onboarding & Learning 🧠

Help new users understand how a product works

Examples:

- Walkthroughs
- Playthroughs

Fields

✓ Use a one col design if possible and minimum 50 px height

✓ Use **Autocomplete**, it will save time when filling out (Affordance)

✓ Use appropriate size per form field for the information needed

✓ Make use of ghost texts, groups and whitespace

✓ Easy understandable optional forms

NAME

Ghost text here

This is a hint text to help user.

ADDRESS (*)

Gerstenweg 1



ZIP-CODE

4061

Buttons

The level of emphasis helps determine its appearance, typography & placement

✓ Use min 50 px height and width

✓ Ghost buttons for secondary actions

✓ For hover use a brighter color

✓ For disabled use transparency eg. 30 % for background, 70 % text

Secondary

Medium emphasis
on the left
lightest color

Cancel

Primary

High emphasis
on the right
darkest color

Save

Drop Downs

Great for more items as they keep the UI minimal

✓ Great for ranges

✓ Often combined with checkboxes or icons or used as filter

✗ Use for less than 3 options

✗ Dropdown width is smaller than form field

SELECT YOUR DISTANCE ^

50 m - 100 m

100 m - 150 m

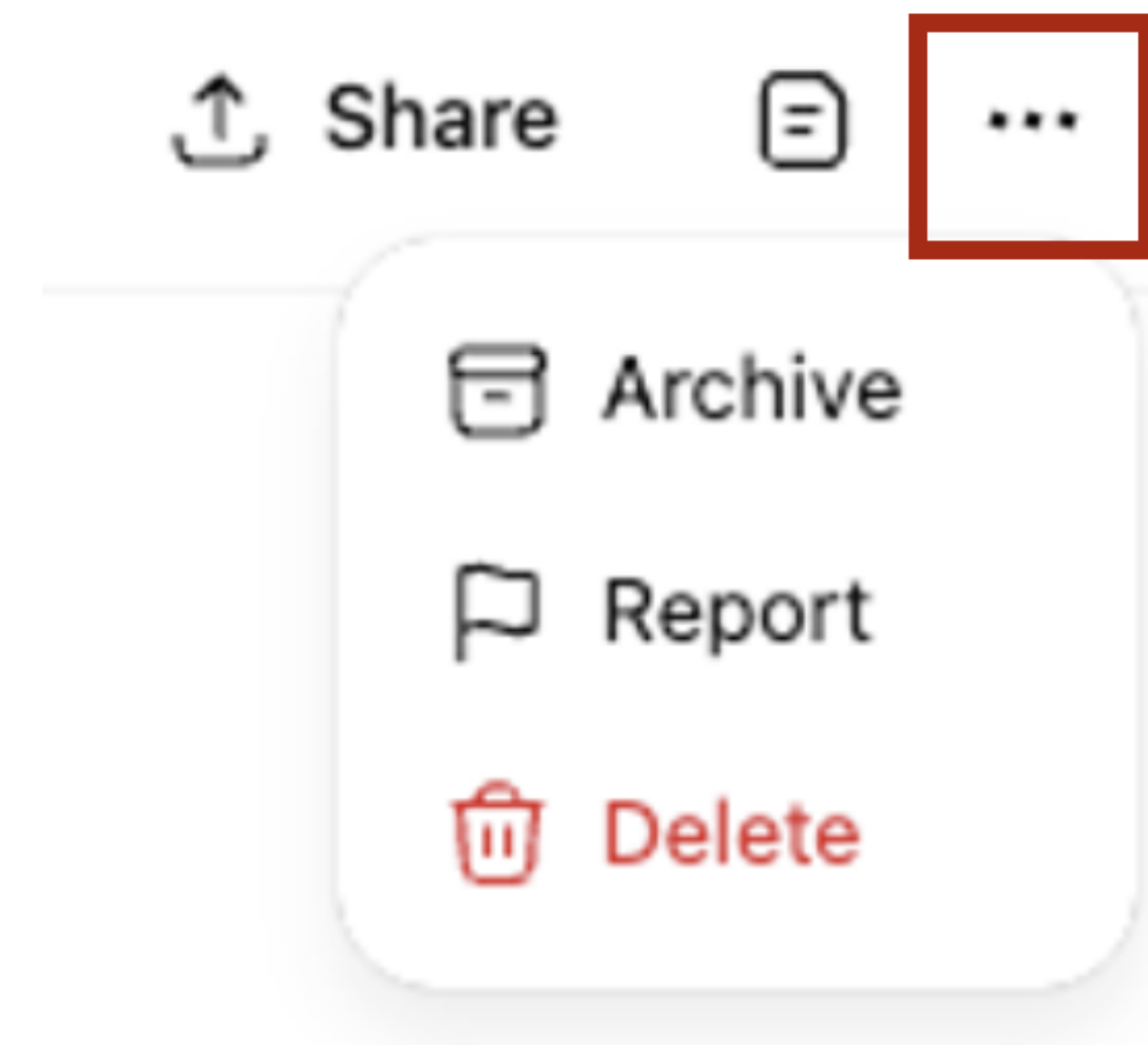
150 m - 200 m

Contextual Menu

**Reduce visual noise,
streamline layouts, and
support focused interaction**

✓ Hide actions that are not primary but that are used enough that they need to be accessible

✗ Hide essential, high-frequency actions behind an extra tap or click eg. stop video

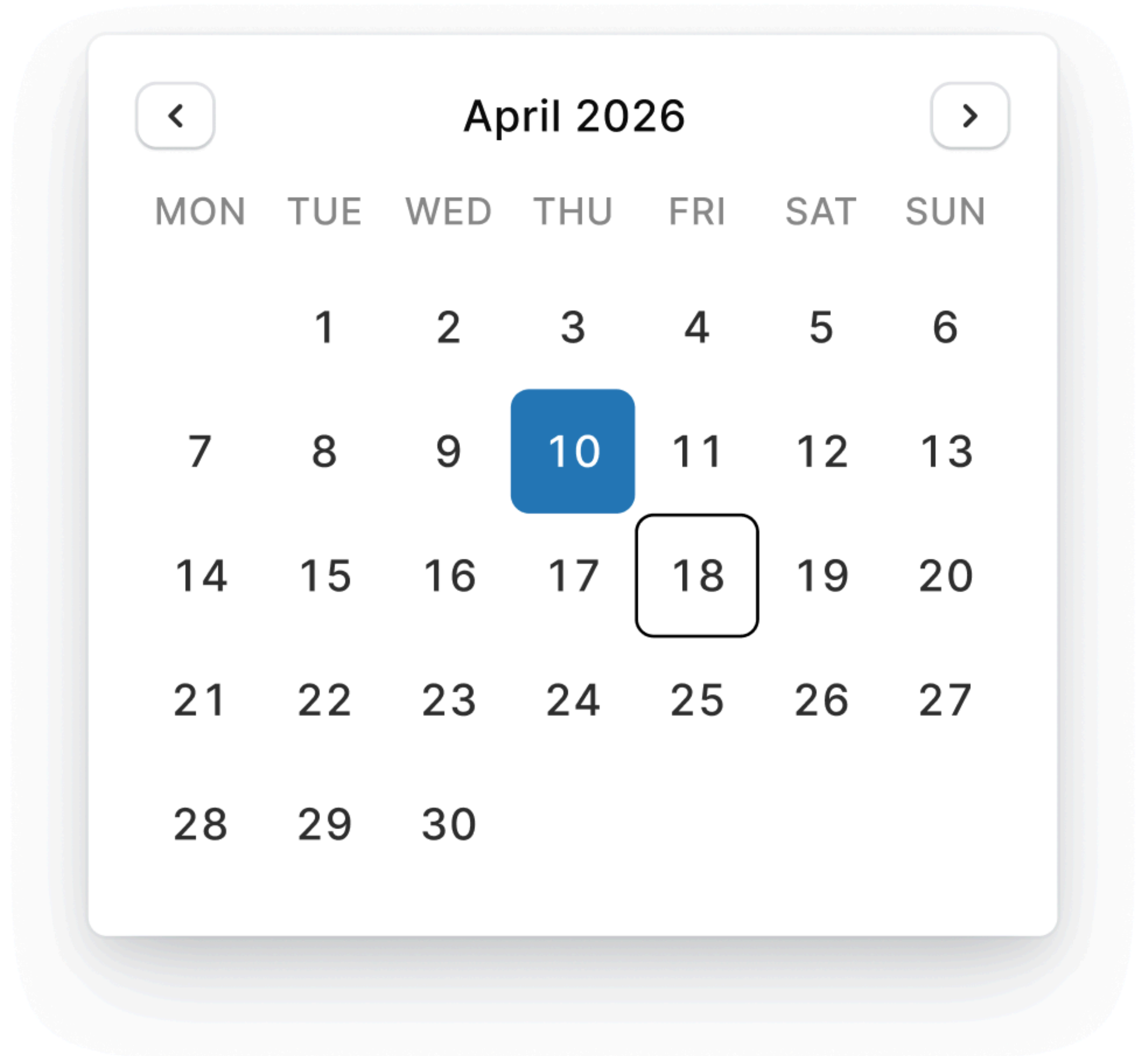


Date Pickers

Used to select date or date range

✓ Forward/backward month, actual date, today and hover should be considered

✓ Combine with a date format field



Radio buttons, Checkboxes

✗ Don't use radio buttons if there are many options

✓ Layout vertically

✓ Click should be on entire space, not only on the element

☒ Remember me

☐ Remember me

☐ Remember me

☒ Remember me

☒ Remember me

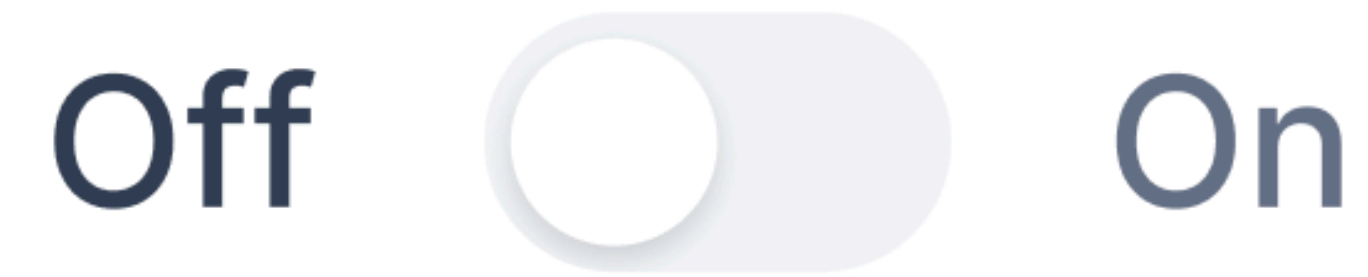
☒ Remember me

Toggle

✓ The color of the text indicates current state

✓ Gives us instant feedback that our action (switching the state) was a success

✗ Don't use toggles for actions that require a "Save" or "Submit" button; they should take effect immediately



Tooltip

✗ Not for mobile devices

✗ Not for much content

✓ Only use if the information is **not** necessary to fulfil a task - otherwise show information

Password guidelines are: use at least 8 characters, include uppercase and lowercase letters, a number, and a special character. Avoid common words or easily guessable information like your name or birthdate.



some more context



Tab bar

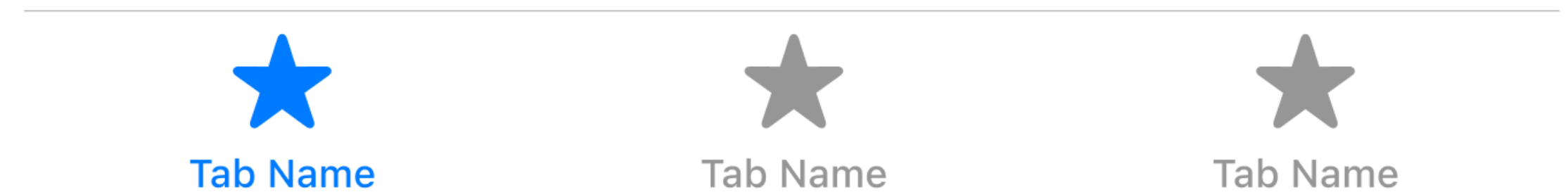
Used for mobile/tablet sides
for navigating to a page

✓ Used on mobile devices

✓ Better than hamburger as it exposes the
information

✓ Fixed at bottom of page

✗ Don't use for more than 5 destinations



Accordion

Showing more information on click

✓ Don't use for more than 10 objects

✗ Try not to add too much child level information

Title ^

Answer the frequently asked question in a simple sentence, a longish paragraph, or even in a list.

Title v

Title v

Title v

Title v

Tab

The most common way to present sections of categories

✓ Make the tab bar at least 90 px tall

✓ Better than accordions

✓ When on mobile use the full horizontal width

Section 1

Section 2

Wizard & Onboarding

Enables untrained or new users to achieve a goal through series of steps

Psychology: if we make the 1st step in the wizard and it is easy to complete the user is more likely to complete all steps

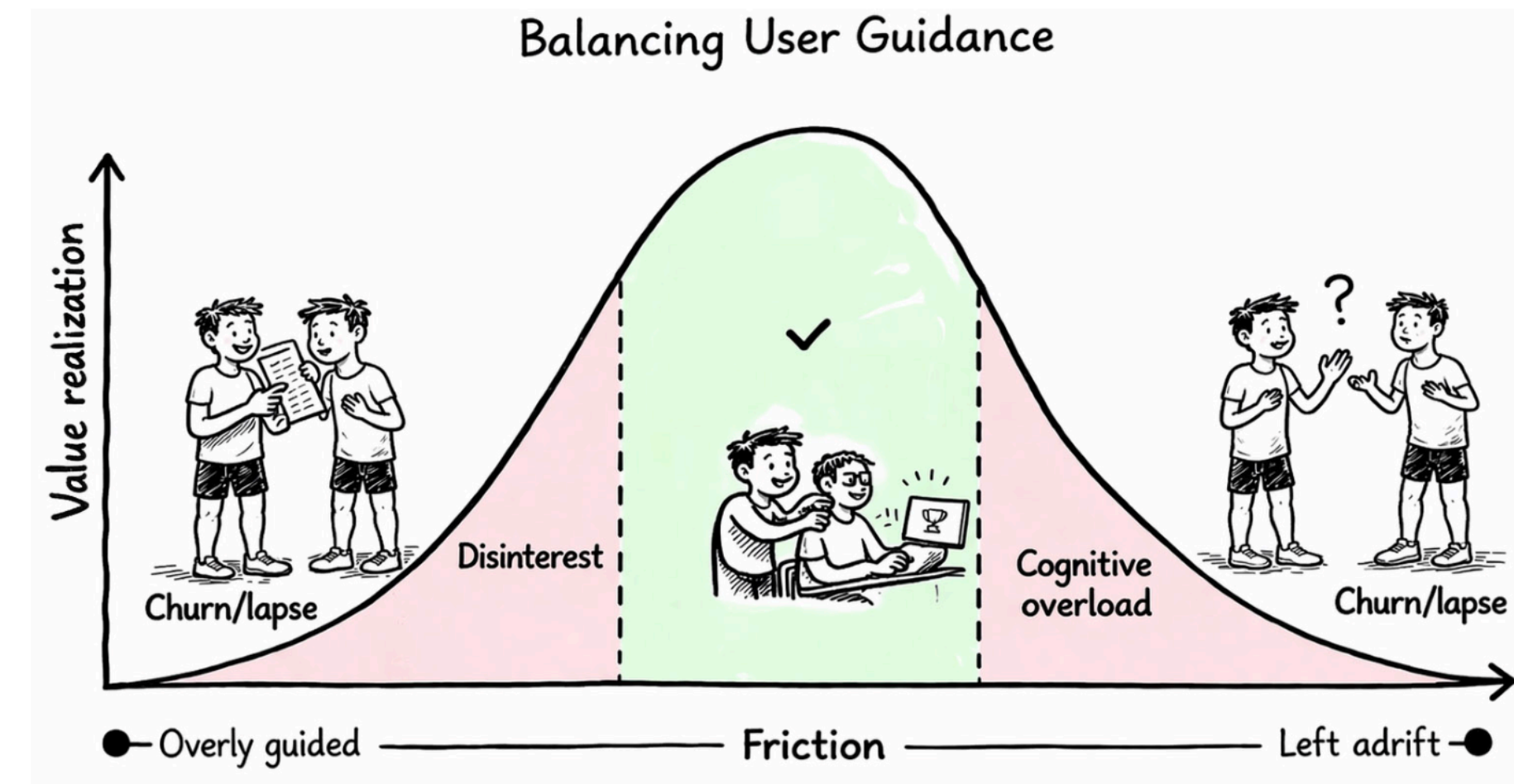
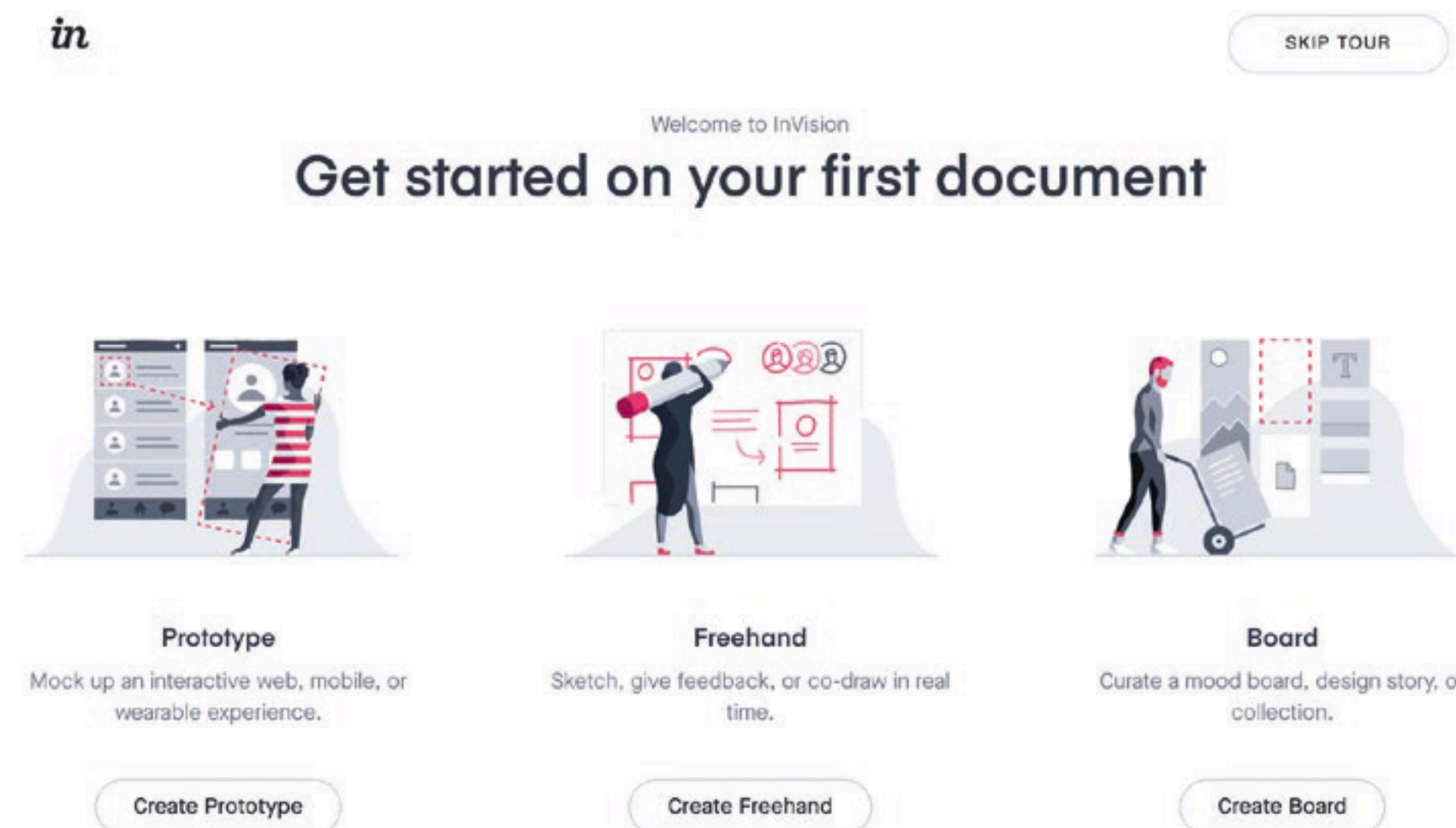
Wizards are great for [onboarding flows](#), where the users need to enter a set of information to start using an application

Example: Username, profile picture



Walkthrough

Guided tour or demo,
presented in a specific order
to introduce complex
workflows or concepts



✓ Good onboarding is not a sequence,
more a balancing act over time.
Challenge grows as users grow.
Guidance fades as confidence builds.

Breadcrumbs

Inform users of their current location, and enable them to quickly move up to a parent level or previous step

✓ Use when you have nested content organized in a hierarchy of two or more levels

[Home](#) / [Primary](#) / [Secondary](#) / Tertiary

Pagination

Show lots of information that cannot be displayed on one screen

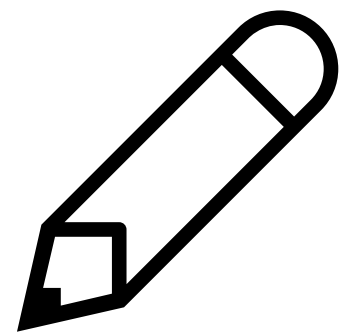
✓ If you can avoid pagination and use infinite scroll do it

✓ Add filter and search function



Icons, Photos

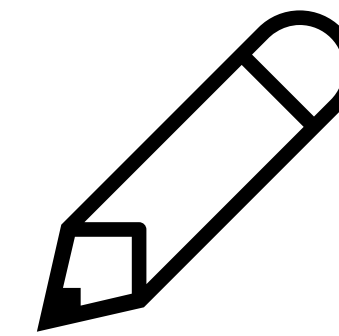
Bridge language barrier and widely understood,
easier to scan/read than text



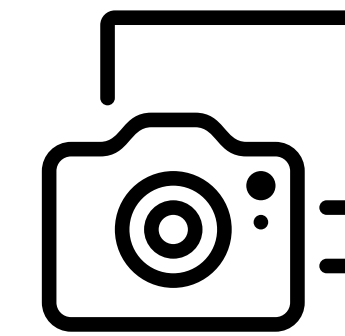
action orientation
ux design side



illustrative orientation
graphical side



universal



needs label

✓ Do not make action icons very illustrative, icons should be very simple, easy to scan, and easy to understand

✓ If you are not sure if a user is going to understand what your icon does or what it is at a glance add a label

Photos

**Photos are more than decoration,
they are tools to communicate**

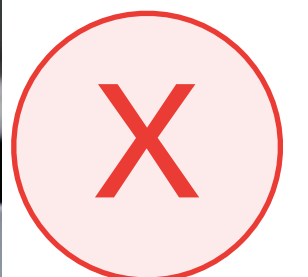
✓ Showing personality rises emotions

✓ Have clear communication that fits your goal

✓ Less is more

✓ Use good quality but avoid long loading time

✓ Beware of colors



Table

A table is a grid of columns and rows, allowing for easy scanning, comparing, and analysing large data sets

✗ Avoid using icons in data tables especially decorative ones

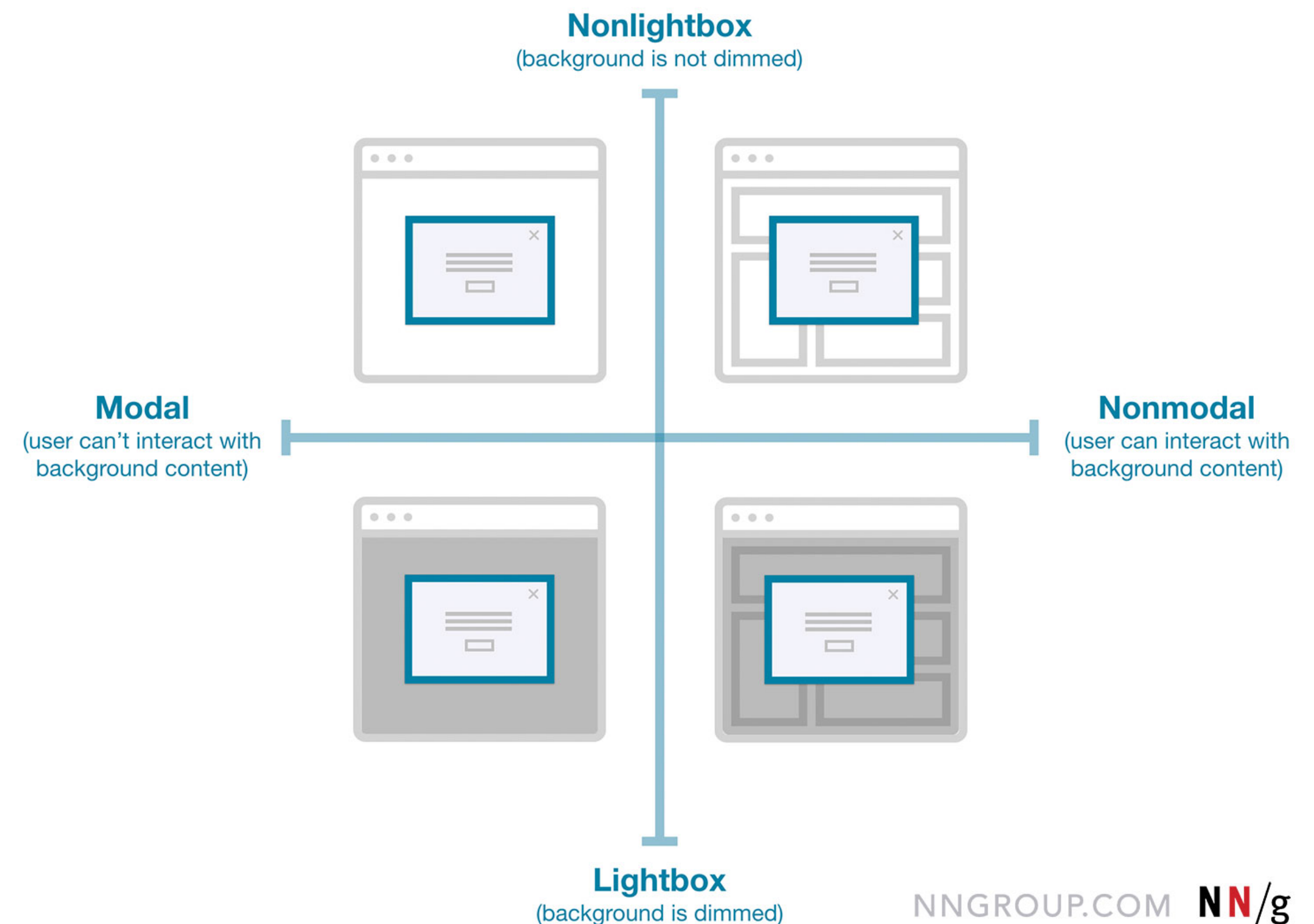
✓ Use zebra striping when you manage large data tables as it

The diagram illustrates a data table with various interactive elements and annotations. The table has a search bar at the top, a header row, and a body with multiple rows. Annotations point to specific features: 'Search' points to the search bar; 'Header row' points to the first row of the table; 'Sorting' points to the 'Date modified' header; 'Rows' points to the data rows; 'Columns' points to the table columns; 'Row checkbox' points to the checkboxes in the first column; and 'Pagination' points to the footer controls.

Search		Header row	
Search product by name			
Product	Category	Price	Sorting ^ Date modified
☐ Avocado toast	Breakfast	\$10.00	02/05/2021
☐ Hotcakes	Breakfast	\$10.00	02/23/2021
☐ Tuna salad	Lunch/Dinner	\$15.00	03/21/2021
☐ Smoothie	Snacks	\$15.00	04/02/2021
☐ Beef stew	Lunch/Dinner	\$25.00	04/17/2021
☐ Granola	Snacks	\$7.00	05/14/2021
Rows per page		25	1-10 of 100 << >>

Modal <> Dialog <> Overlay <> Lightbox

- ✓ **Dialog**: „Conversation“ between user <> system
- ✓ **Overlay** (or Popup): appears on top of page content
- ✓ **Modal**: user must interact with overlay + background disabled, for a single, self contained task; or to warn users about potential mistakes
- ✓ **Nonmodal**: user must interact with overlay + background enabled to work with
- ✓ **Lightbox**: dimmed background to focus on modal



Modal <> Dialog <> Overlay <> Lightbox

If you need to	Use	
Force a decision	Modal	X Avoid unnecessary modals! It's a UX 😞
Ask for user input	Dialog that blocks background	
Show context information	Overlay with non blocking info	X Often overlays appear at wrong time, interrupt users during critical tasks, use poor language
Display media	Lightbox with dimmed background	

Top 10 Application-Design Mistakes

1. Poor Feedback
2. Inconsistency
3. Bad Error Messages
4. No Default Values
5. Unlabeled icons
6. Hard-to-Acquire Targets
7. Overuse of Modals
8. Meaningless Information
9. Junk-Drawer Menus
10. Proximity of Destructive and Confirmation Actions

Dark Patterns

Dark Patterns

Problematic patterns, such as:

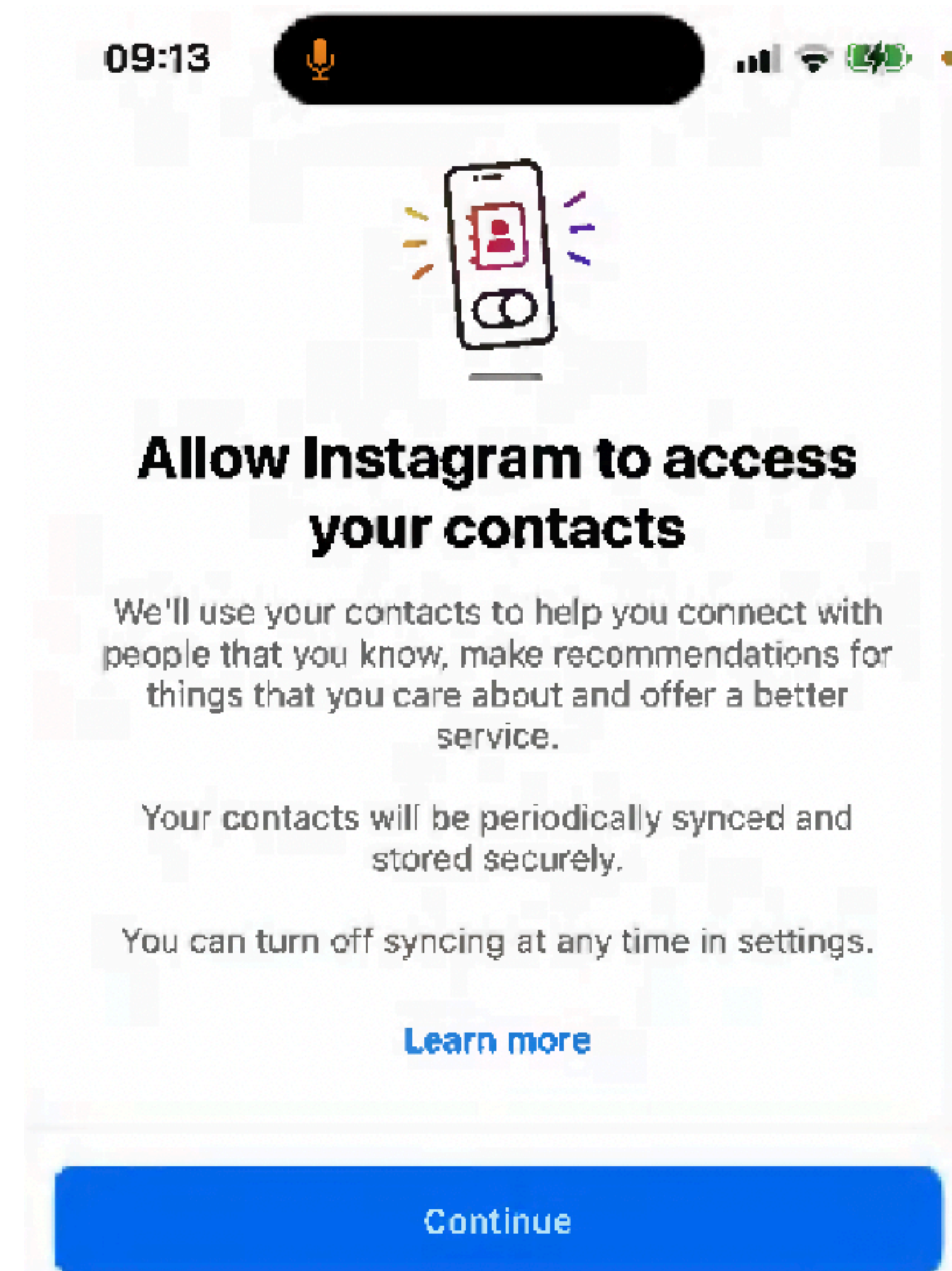
- ✓ Manipulative designs
- ✓ Confusing navigation
- ✓ Excessive pop-ups



Dark Patterns

Example Instagram:

Which makes use of the forced action pattern by providing no alternative option but to allow access to private data in order to proceed



Hands on

- ✓ Have a look at the instagram example on the slide
- ✓ Identify the Anti-Patterns (Discuss as a group of 2)
- ✓ How would you redesign them?

Afterwards: Explain your findings in plenum

Challenge #3

**UI-Pattern classification
& complex components**

2 people per team

🔍 Have a look at the examples from Challenge #1, #2, or Levis

👁 Identify 5 UI-Design Patterns in the interface
use ui-patterns.com and the slides

- ★ Write down:
- Name and explanations of pattern
 - Pattern category/classification
 - UX Law
 - Visually mark pattern on interface
 - Heuristic

Afterwards: Redesign the Pattern with Figma as complex component

👉 Challenge #3

Pattern: Chunking

Pattern Category: Perception and memory / Comprehension

When you enter a phone number into your iPhone, the phone number is automatically chunked into smaller bits of information that makes it easier for the user to read and remember. Breaking down the full phone number into more chunks makes the number easier to remember.

Heuristic: #6 Recognition rather than recall



UX Laws & Persuasive Design

UX Laws

Psychology-based strategies that deliberately direct users toward intended outcomes

✓ Derived from behavioral psychology and cognitive science

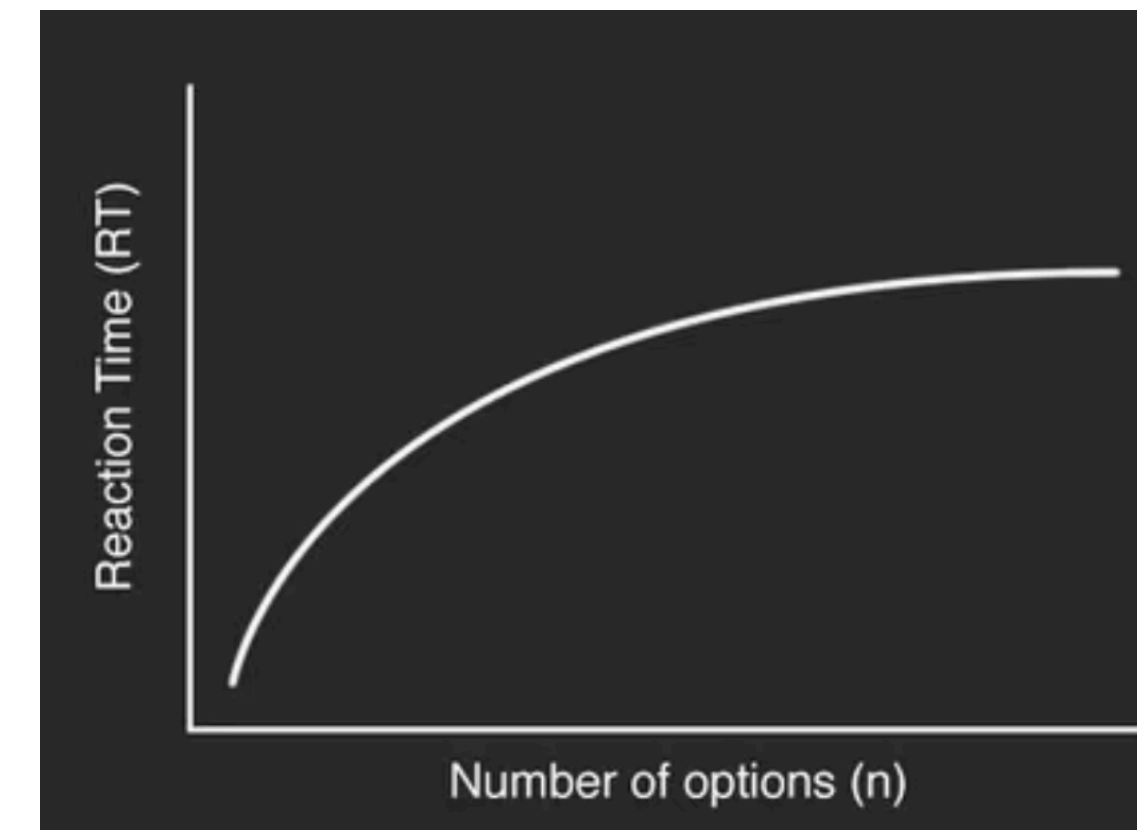


Hick's law

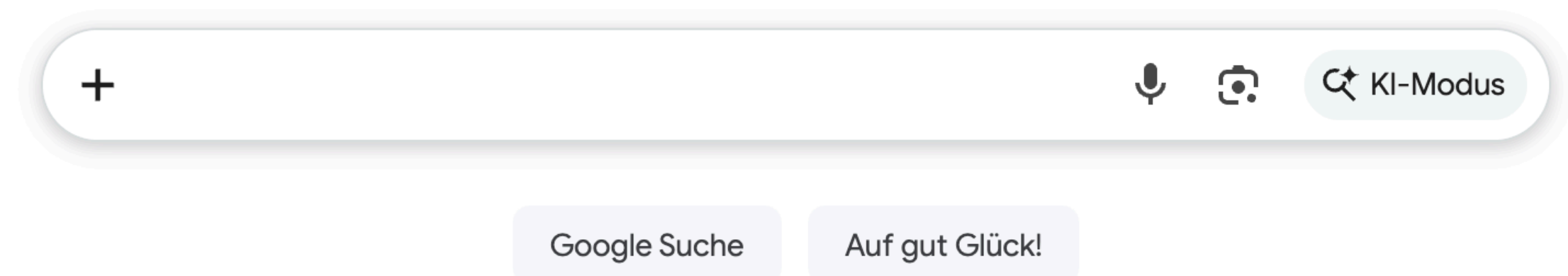
The more stimuli (or choices) users face, the longer it will take them to make a decision

- ✓ The number of possible choices should be limited.
- ✓ Having too many things to choose from can cause disorientation for the user

Example: Google keeps the decisions required to enter a keyword to a minimum by eliminating any additional content



$$RT = a + t * \log_2(n)$$



Tesler's law

For any system there is a certain amount of complexity that cannot be reduced

✓ Also known as the „law of conservation of complexity“



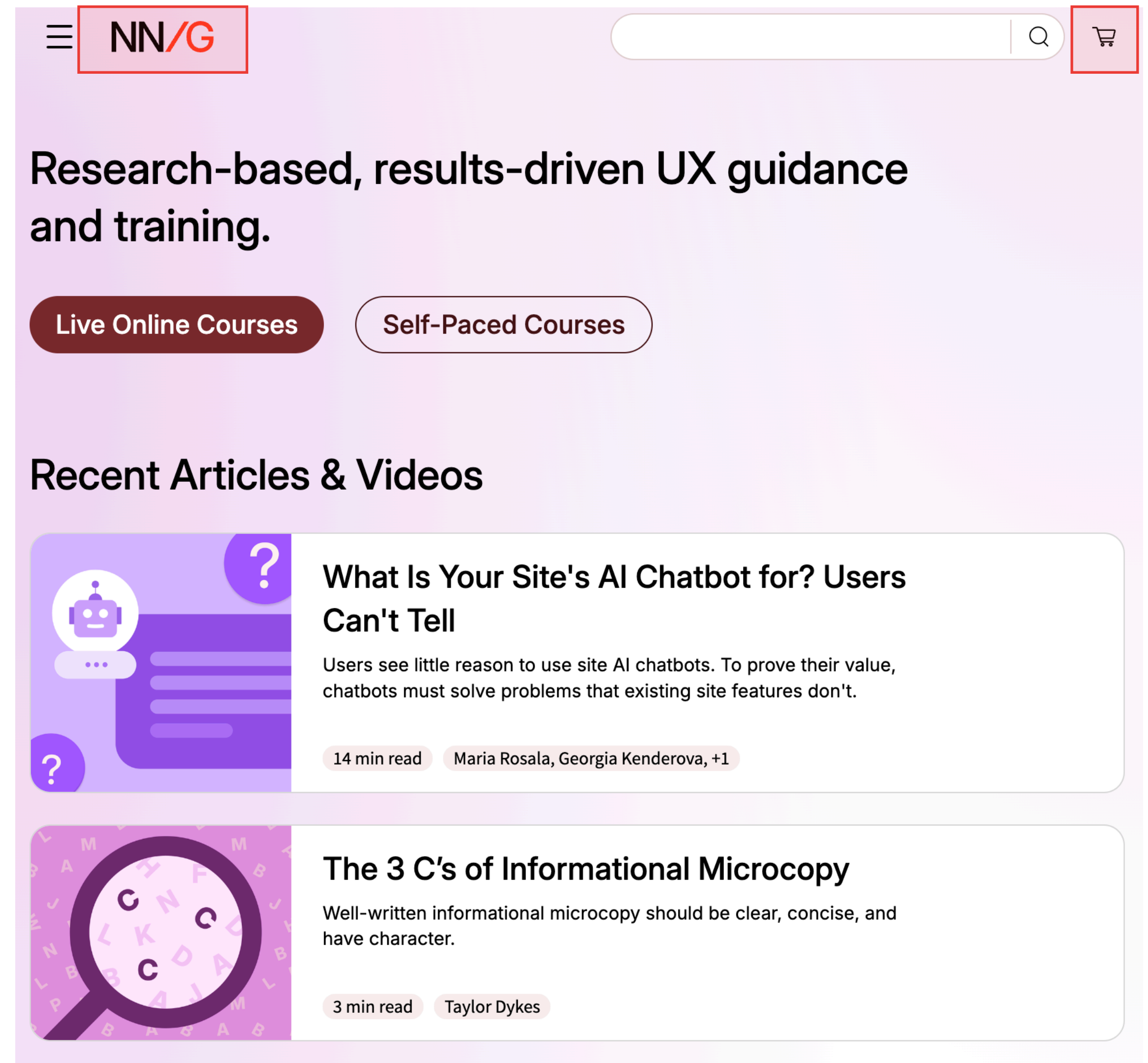
✓ **Progressive disclosure** (interaction design technique) means to display only important actions or content by default. Simplify your design by hiding complexity.

Jakob's law

Users expect that websites work the same way as other websites

✓ Faster decisions when using established patterns

Example: Think of the logo which is usually on top of a page or the shopping chart icon



Persuasive Design

Crafting products, that not just to look good, but that nudge users toward specific behaviors

Eg. Call-to-Action-Buttons, Social Proof, FOMO

„10,000 people bought this today“

„Recommended by doctors“



SIXT
RENT A CAR

BIS ZU
15%
RABATT

**NICHT WARTEN.
JETZT BUCHEN.**

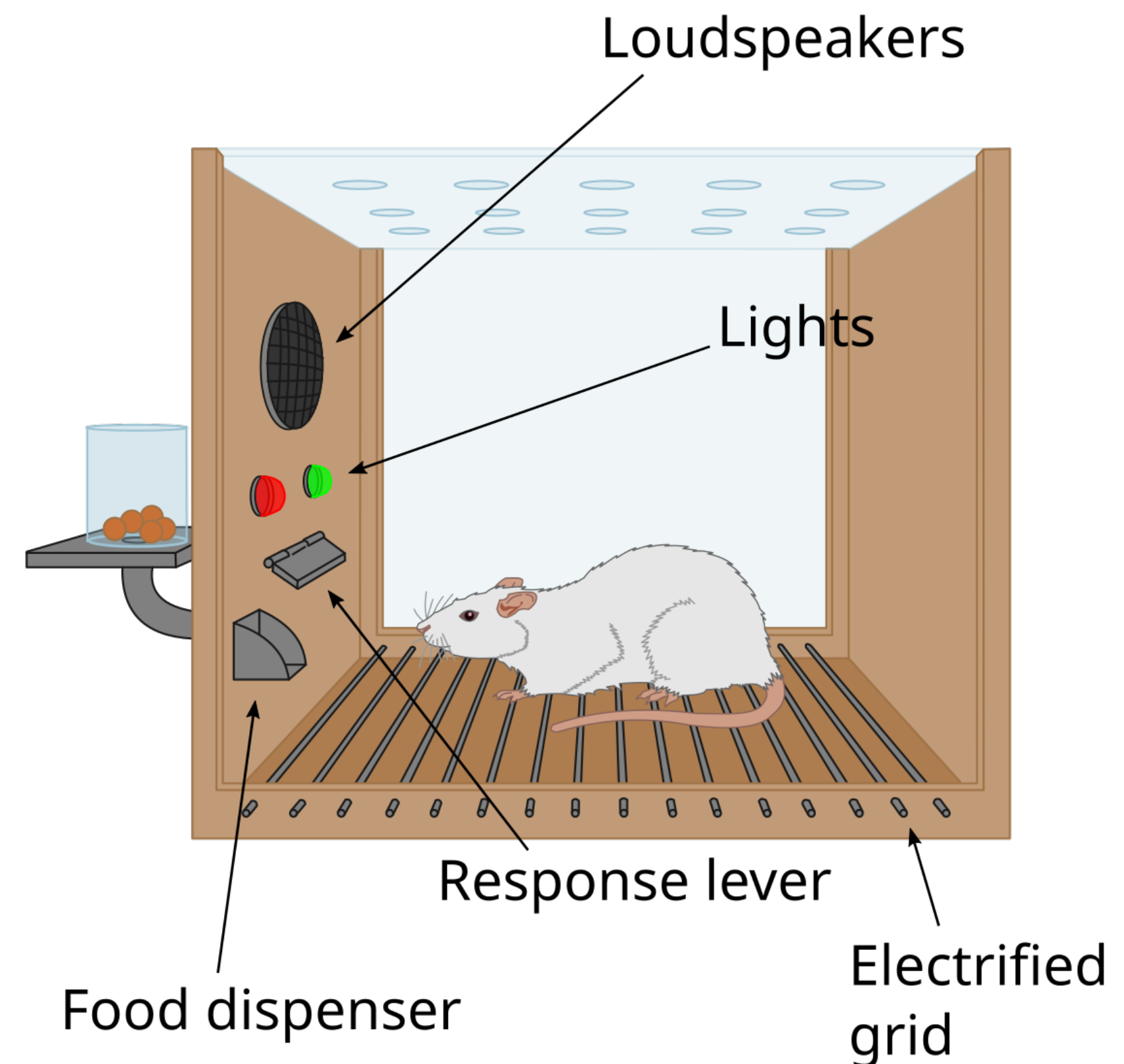
Die Uhr wird vorgestellt, und damit bleibt Ihnen eine Stunde weniger Zeit, sich **bis zu 15% Rabatt** auf Ihre nächste Miete zu sichern.*
Das Angebot gilt nur bis zum 31. März.

JETZT BUCHEN

How Technology Shapes behavior: Skinner

Skinner studied how the behavior of animals could be shaped by teaching them to perform desired actions in response to specific stimuli in an isolated environment

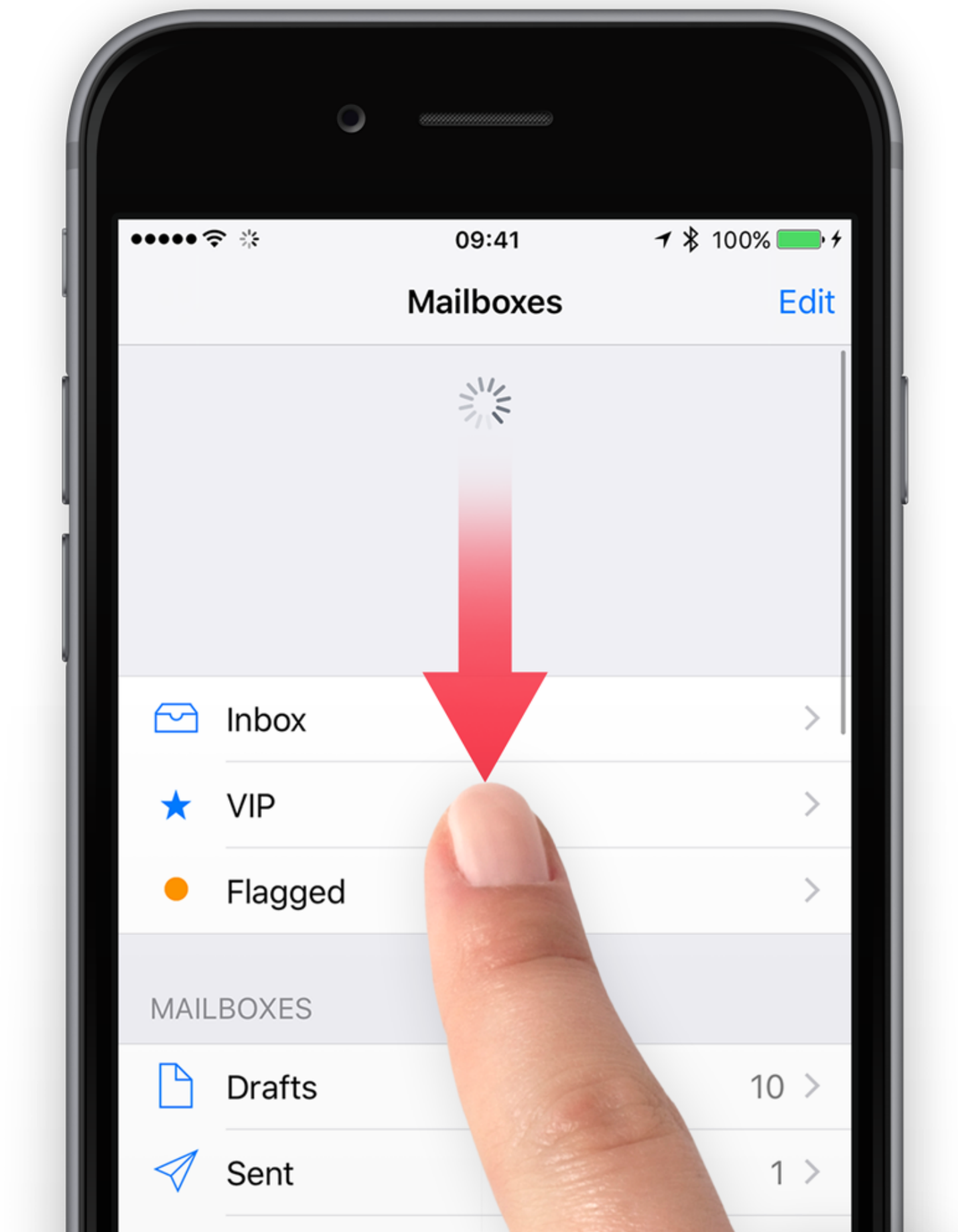
✓ operant conditioning



How Technology Shapes behavior: Skinner

Digital products and services employ various methods with the goal of shaping human behaviour, we see many of the interaction patterns in apps we use every day:

- **Pull to refresh:** similar to “lever pressing”
- **Infinite loops:** continuous search for next reward
- **Social affirmation:** likes, comments
- **Personalised content:** possibility of receiving rewards



References

Moser, C., User Experience Design: Mit Erlebniszentrierter Softwareentwicklung zu Produkten die Begeistern, Springer Verlag, 2012

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Staiano, F., Designing and Prototyping Interfaces with Figma Learn essential UX/UI design principles by creating interactive prototypes for mobile, tablet, and desktop, 2022

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Staiano, F., Designing and Prototyping Interfaces with Figma Learn essential UX/UI design principles by creating interactive prototypes for mobile, tablet, and desktop, 2022

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